

NTC - OHBN2

Operation Guide

SPTI

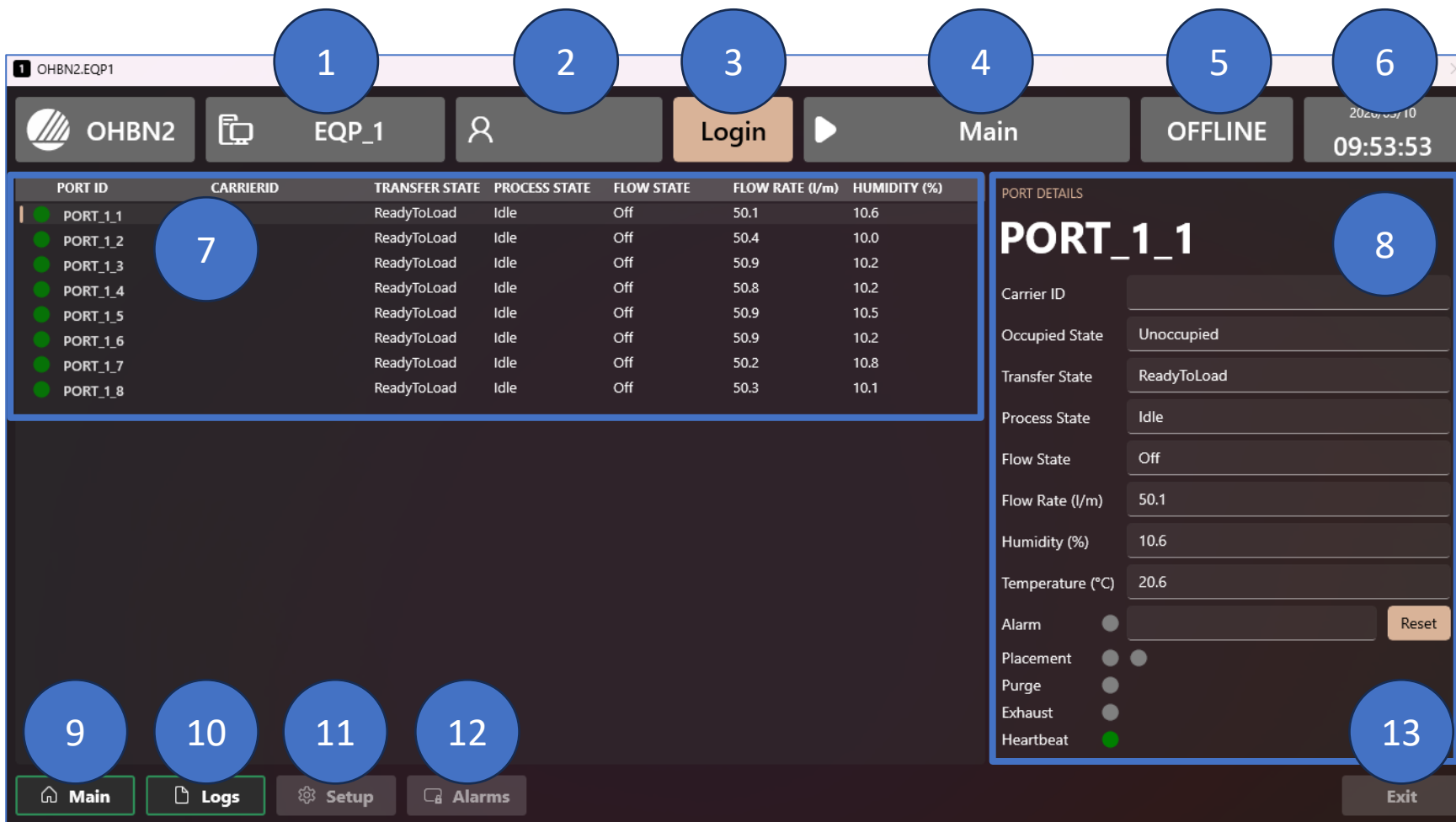
2026-03-10

Outline

- GUI 介紹
- 使用者登入
- 機台設定
- SECS / GEM 設定
- N-Port 設定
- Process 設定
- 權限設定
- 充氣操作流程

GUI 介紹

GUI 介紹



標題	說明
1	機台名稱
2	使用者登入帳號
3	使用者登入&登出
4	當前頁面位置
5	SECS Control State
6	當前時間
7	LoadPort 列表 綠燈：正常運行 黃燈：Out of Service 紅燈：Alarm
8	LoadPort 狀態
9	主畫面
10	日誌查詢
11	機台設定
12	告警查詢
13	關閉程式

使用者登入 / 登出

使用者登入 / 登出 (1/2)

OHBN2.EQP1

OHBN2

EQP_1

Login

Main

OFFLINE

2026/03/10
13:37:32

PORT ID	CARRIERID	TRANSFER STATE	PROCESS STATE	FLOW STATE	FLOW RATE (l/m)	HUMIDITY (%)
PORT_1_1		ReadyToLoad	Idle	Off	51.0	10.8
PORT_1_2		ReadyToLoad	Idle	Off	50.0	10.3
PORT_1_3		ReadyToLoad	Idle	Off	50.8	10.1
PORT_1_4		ReadyToLoad	Idle	Off	50.7	10.2
PORT_1_5		ReadyToLoad	Idle			
PORT_1_6		ReadyToLoad	Idle			
PORT_1_7		ReadyToLoad	Idle			
PORT_1_8		ReadyToLoad	Idle			

Sign In

Username
admin

Password
admin

Sign InCancel

PORT DETAILS

PORT_1_1

Carrier ID

Occupied State
Unoccupied

Transfer State
ReadyToLoad

Process State
Idle

Flow State
Off

Flow Rate (l/m)
51

Humidity (%)
10.8

Temperature (°C)
20.2

Alarm

Placement

Purge

Exhaust

Heartbeat

Reset

Main

Logs

Setup

Alarms

Exit

使用者登入 / 登出 (2/2)

The screenshot displays the OHBN2.EQP1 control interface. At the top, a navigation bar includes the OHBN2 logo, a monitor icon labeled 'EQP_1', a user profile icon labeled 'admin', a 'Logout' button (highlighted with a blue box), a play button labeled 'Main', an 'OFFLINE' status indicator, and a timestamp '2026/03/10 13:30:39'.

Below the navigation bar is a table with the following columns: PORT ID, CARRIERID, TRANSFER STATE, PROCESS STATE, FLOW STATE, FLOW RATE (l/m), and HUMIDITY (%). The table lists eight ports, all with a 'ReadyToLoad' transfer state and 'Idle' process state.

A 'Sign Out' dialog box is centered on the screen, asking 'Are you sure?' with 'Sign Out' and 'Cancel' buttons.

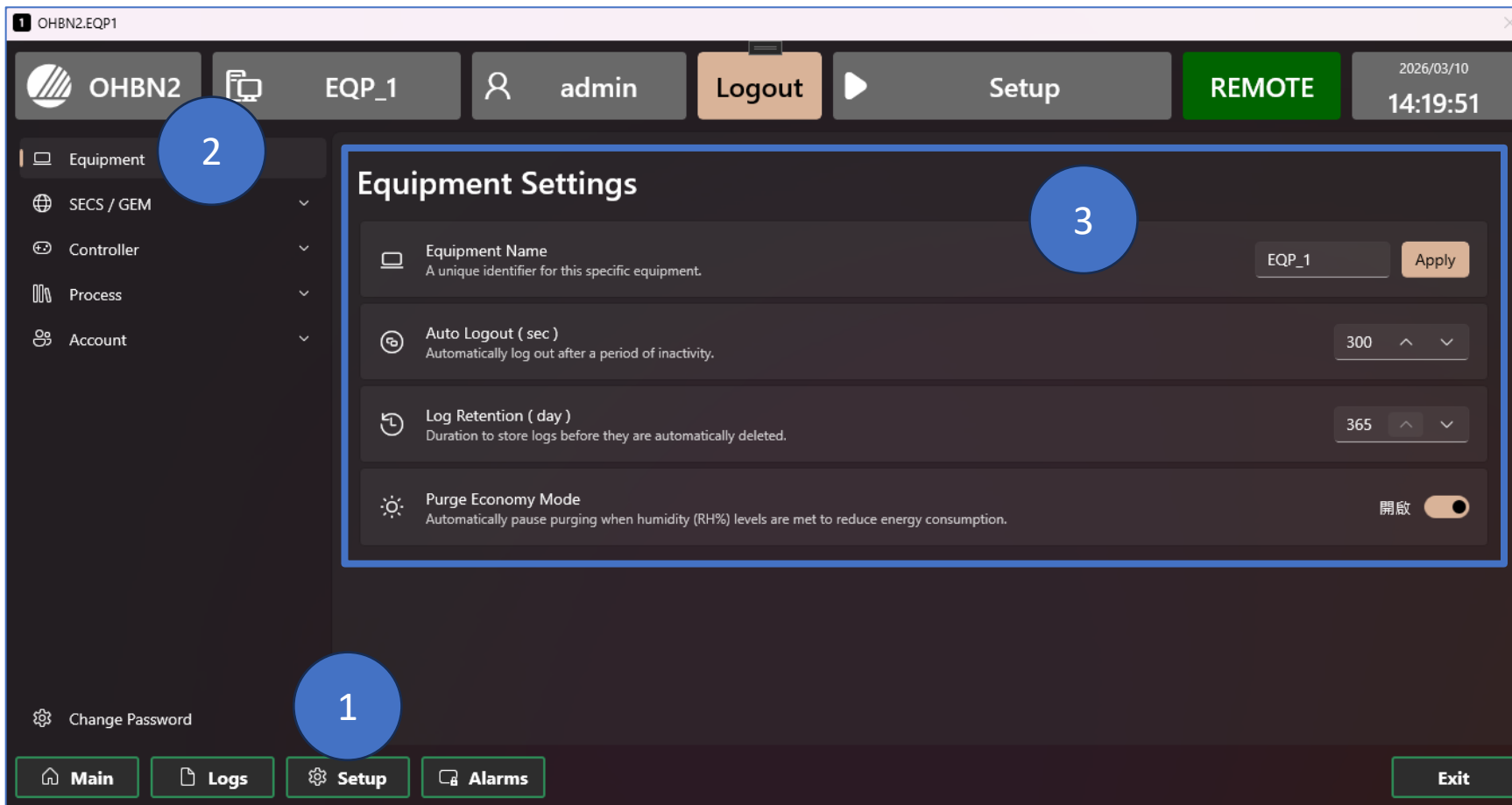
On the right side, the 'PORT DETAILS' for 'PORT_1_1' are shown, including fields for Carrier ID, Occupied State (Unoccupied), Transfer State (ReadyToLoad), Process State (Idle), Flow State (Off), Flow Rate (l/m) (50), Humidity (%) (10.4), and Temperature (°C) (20.6). There is also an 'Alarm' section with a 'Reset' button and a 'Heartbeat' indicator.

At the bottom, a navigation bar contains buttons for 'Main', 'Logs', 'Setup', 'Alarms', and 'Exit'.

PORT ID	CARRIERID	TRANSFER STATE	PROCESS STATE	FLOW STATE	FLOW RATE (l/m)	HUMIDITY (%)
PORT_1_1		ReadyToLoad	Idle	Off	50.0	10.4
PORT_1_2		ReadyToLoad	Idle	Off	50.0	11.0
PORT_1_3		ReadyToLoad	Idle	Off	50.1	10.7
PORT_1_4		ReadyToLoad	Idle	Off	50.5	10.1
PORT_1_5		ReadyToLoad	Idle	Off	50.1	10.1
PORT_1_6		ReadyToLoad	Idle	Off	50.9	10.4
PORT_1_7		ReadyToLoad	Idle	Off		
PORT_1_8		ReadyToLoad	Idle	Off		

機台設定

機台設定



標題	說明
1	點選 Setup
2	點選 Equipment
3	Equipment Name: 機台名稱 Auto Logout: 自動登出時間 Log Retention: Log 保存天數 Purge Economy Mode: 節能模式

SECS / GEM 設定

SECS / GEM 參數設定 (1/2)

OHBN2.EQP1

OHBN2 EQP_1 admin Logout Setup REMOTE 2026/03/10 14:23:58

Equipment SECS / Option Change State Controller Process Account

SECS / GEM Options

Device ID
A unique 15-bit identifier (0 - 32767) used by the Host to identify this specific equipment in the SECS/GEM network. 0

Is Active
Defines whether the equipment acts as the Active or Passive entity in establishing the HSMS TCP/IP connection. 關閉

Host
The target IP address of the Host (EAP/MCS) used for HSMS communication. 127.0.0.1

Port
Specifies the TCP/IP port number (0 - 65535) used for HSMS communication. 5168

T3 (sec)
Reply Timeout: Max time to wait for a response message before declaring a timeout. 45

T5 (sec)
Connect Separate Timeout: The time to wait between connection attempts when in Active mode. 10

Change Password

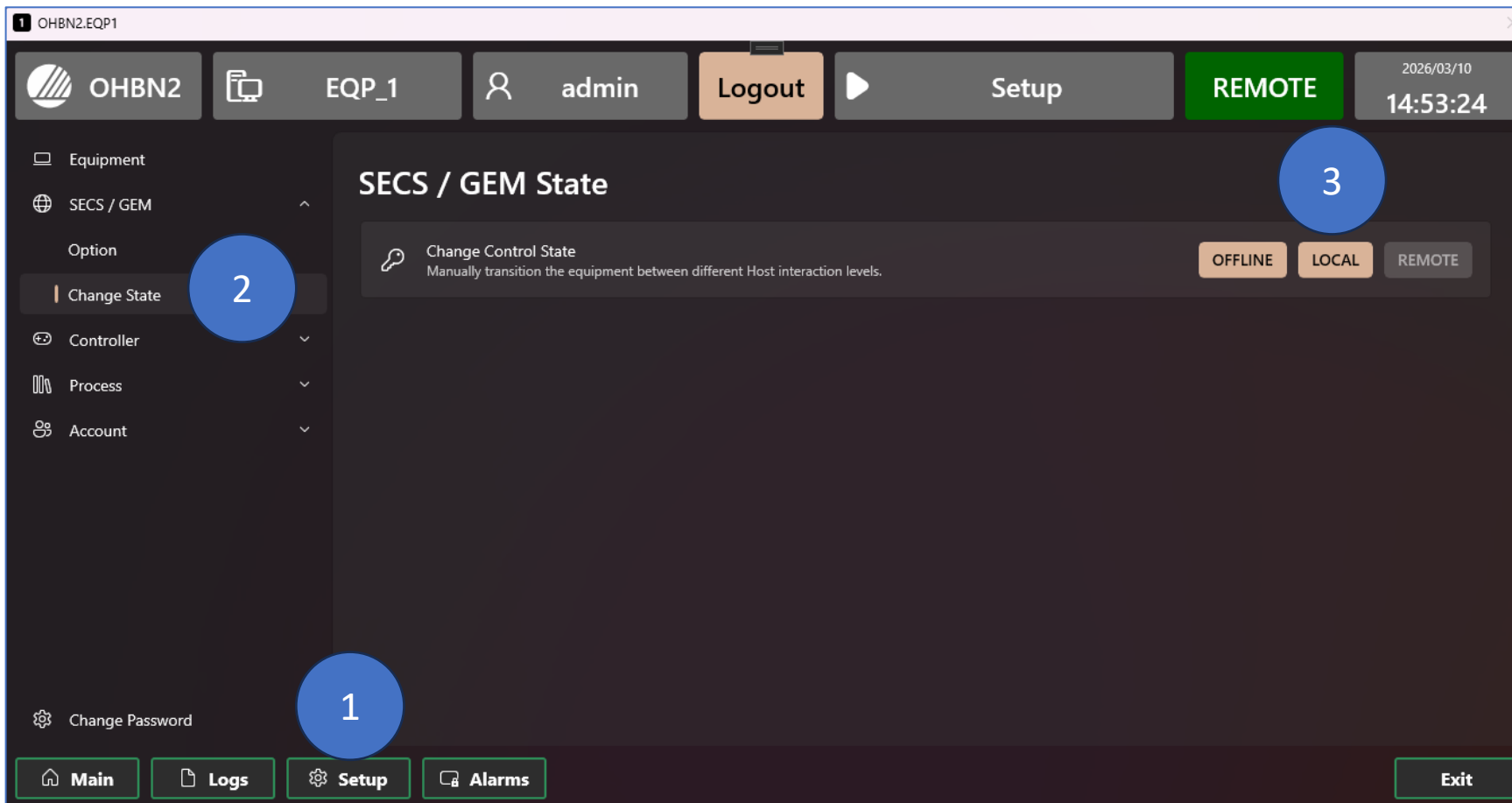
Main Logs Setup Alarms Exit

標題	說明
1	點選 Setup
2	點選 Option
3	SECS / GEM 設定

SECS / GEM 參數設定 (2/2)

SECS / GEM Options	標題	說明
 Device ID A unique 15-bit identifier (0 - 32767) used by the Host to identify this specific equipment in the SECS/GEM system.	Device ID	設備識別碼 (SECS 通訊位址)
 Is Active Defines whether the equipment acts as the Active or Passive entity in establishing the HSMS TCP/IP connection.	IsActive	設定連線模式為 主動 (Active) 或 被動 (Passive)
 Host The target IP address of the Host (EAP/MCS) used for HSMS communication.	Host	IsActive 為 true 時，是遠端設備 IP IsActive 為 false 時，是本機端監聽 (Bind) 的 IP
 Port Specifies the TCP/IP port number (0 - 65535) used for HSMS communication.	Port	IsActive 為 true 時，是遠端 TCP Port IsActive 為 false 時，是本機端監聽 (Bind) 的 TCP Port
 T3 (sec) Reply Timeout: Max time to wait for a response message before declaring a timeout.	T3	發送 Primary 訊息後等待 Secondary 訊息回應的時限
 T5 (sec) Connect Separate Timeout: The time to wait between connection attempts when in Active mode.	T5	TCP 連線建立後，等待 HSMS 狀態轉換至 Connected 的時限
 T6 (sec) Control Timeout: Max time to wait for a response to a control message (e.g., Select/Deselect).	T6	發送控制訊息 (如 Select.req) 後等待回應 (Select.rsp) 的時限
 T7 (sec) Not Selected Timeout: Max time a TCP connection can remain open without an HSMS session being selected.	T7	等待 Select 過程完成以進入 Selected 狀態的時限
 T8 (sec) Network Timeout: Max time allowed between characters within a single HSMS message.	T8	接收單一訊息時，各個資料區段 (Chunk) 之間的時間間隔
 Link Test Interval (sec) The interval between Linktest integrity checks when the connection is idle (0 to disable).	LinkTestInterval	設定 LinkTest 請求之間的時間間隔
 Communication State Specifies the initial Communicate State (Disabled, Enabled) upon equipment startup.	Control State	初始狀態為 OffLine or OnLine
 Control State Specifies the initial Control State (OffLine or OnLine) upon equipment startup.	Control OffLine Sub-State	初始狀態 OffLine 後，進入下一個狀態 (EquipmentOffLine or HostOffLine or AttemptOnLine)
	Control Attempt OnLine Failure Sub-State	嘗試連線失敗後，進入下一個狀態狀態 (EquipmentOffLine or HostOffLine)
	Control OnLine Sub-State	初始狀態為 OnLine 後，進入下一個狀態 (Local or Remote)

SECS / GEM 切換狀態



標題	說明
1	點選 Setup
2	點選 Change State
3	切換 SECS / GEM 狀態

N-Port 設定

N-Port 設定 (1/2)

OHBN2.EQP1

OHBN2

EQP_1

admin

Logout

Setup

OFFLINE

2026/03/10 11:21:41

Equipment

SECS / GEM

Controller

NPort

Process

Account

Change Password

1

NPort

1

2

3

3

Host

127.0.0.1

4

Is Available

開啟

Channel

1

Port

4001

Channel

2

Port

4002

LoadPorts

PORT_1_1

Port No (Must be unique for SECS)

1

Name

Update

Main

Logs

Setup

Alarms

Exit

標題	說明
1	點選 Setup
2	點選 NPort
3	N-Port 列表
4	N-Port 設定

N-Port 設定 (2/2)

Host
127.0.0.1

Is Available
☒ 開啟

Channel Port
1 4001

Channel Port
2 4002

LoadPorts
PORT_1_1

Port No (Must be unique for SECS)
1

Name
PORT_1_1

Is Available
☒ 開啟

Is Enabled
☒ 開啟

Settings saved successfully.

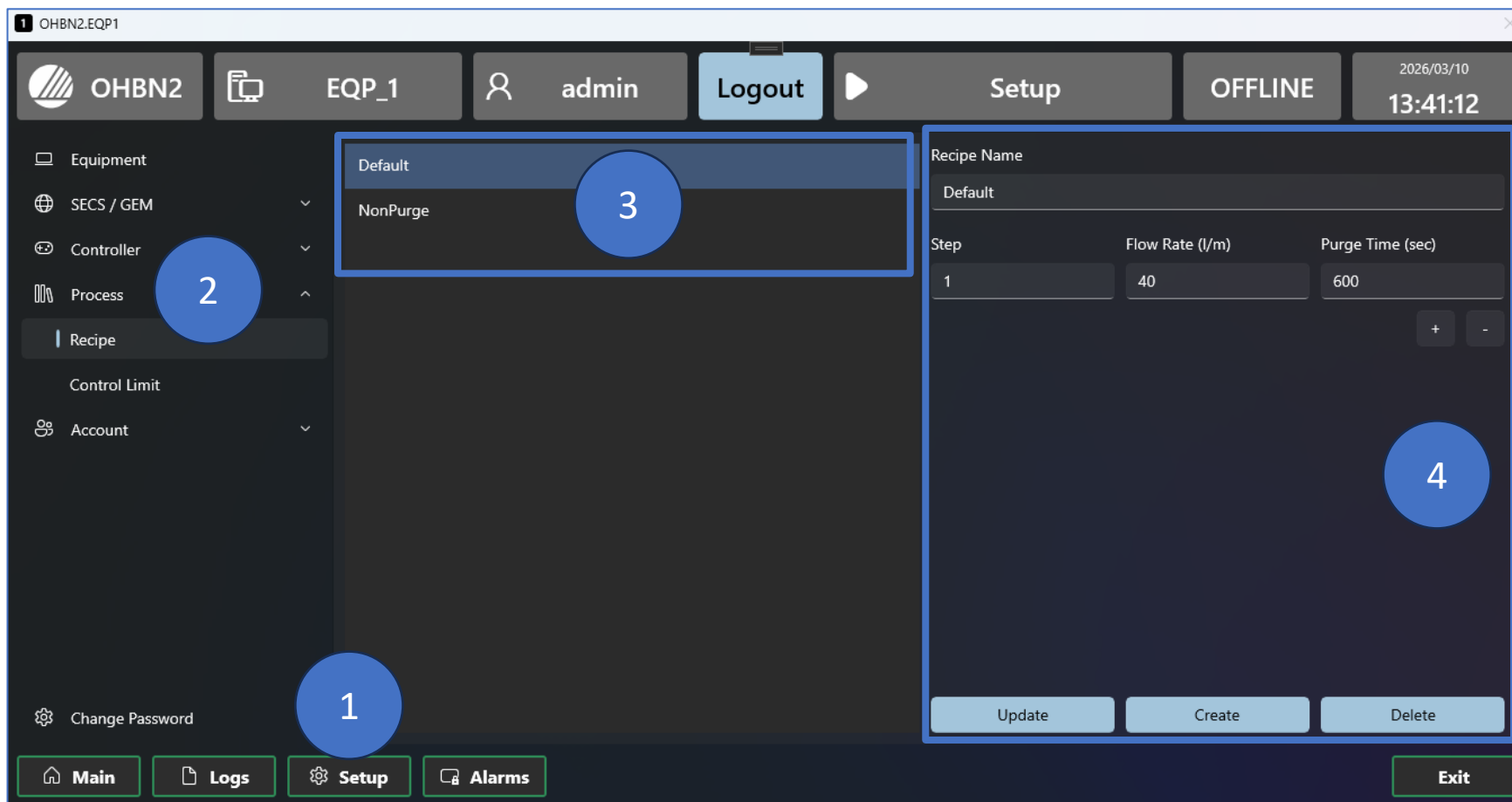
Update

標題	說明
Host	N-Port IP
Is Available	N-Port 可用 / 不可用
Channel 1	PLC 使用的 TCP Port
Channel 2	RFID 使用的 TCP Port
LoadPorts	一個 N-Port 最多連接 8 個 LoadPort。 若要使用，點選每一個 LoadPort 進行設定。
Port No	依照排序 1...8 填寫
Port Name	LoadPort 名稱
Port Is Available	LoadPort 可用 / 不可用
Port Is Enabled	LoadPort 啟用 / 不啟用

修改完後，點選 Update，必須重啟 GUI

Process 設定

流量設定 (1/2)



標題	說明
1	點選 Setup
2	點選 Recipe
3	Recipe 列表 Default: 預設充氣 Recipe (不可刪除) NonPurge: 預設不充氣 Recipe (不可刪除)
4	Recipe 設定

流量設定 (2/2)

Recipe Name

Default 1

Step Flow Rate (l/m) 2 Purge Time (sec) 3

1 40 600

+ -

Settings saved successfully.

Update Create Delete

標題	說明
1	Recipe 名稱
2	流量
3	充氣時間

管制界限

Control Limit Settings



FOUP On Duration (sec)

Specifies the duration (in seconds) that a FOUP remains on the Load Port before triggering a status update or timeout alert.

1

^

▼



Auto Purge Timeout (sec)

Defines the idle time threshold (in seconds) before the system automatically initiates the N2/Dry Air purge sequence to maintain FOUP cleanliness when the FOUP is not in process.

30

^

▼



Low Flow Rate (l/m)

Sets the minimum gas flow rate (L/min) required to maintain internal positive pressure and prevent particle re-entry after the initial high-flow purge.

40

^

▼



Economy Flow Rate (l/m)

Defines the power-saving gas flow rate (L/min) used during extended idle periods or standby mode to minimize N2 consumption while maintaining basic cleanliness.

41

^

▼



Humidity Target (%)

Sets the target relative humidity percentage (RH%) for the FOUP interior. The system will adjust gas flow rates to reach and maintain this moisture level.

5

^

▼



Start Purge Humidity Check Delay (sec)

Sets the delay (in seconds) before the system begins monitoring humidity levels after the purge starts, ensuring the sensor stabilizes and provides an accurate reading.

300

^

▼



Rewetting

Manages the environment recovery process after FOUP removal to restore ambient humidity levels.

▼



Idle Purge

Periodic cleaning cycle when no FOUP is detected.

▼



Flow Rate Tolerance Percentage (%)

Defines the allowable deviation percentage between the actual gas flow and the setpoint. An alarm will be triggered if the flow rate remains outside this range.

100

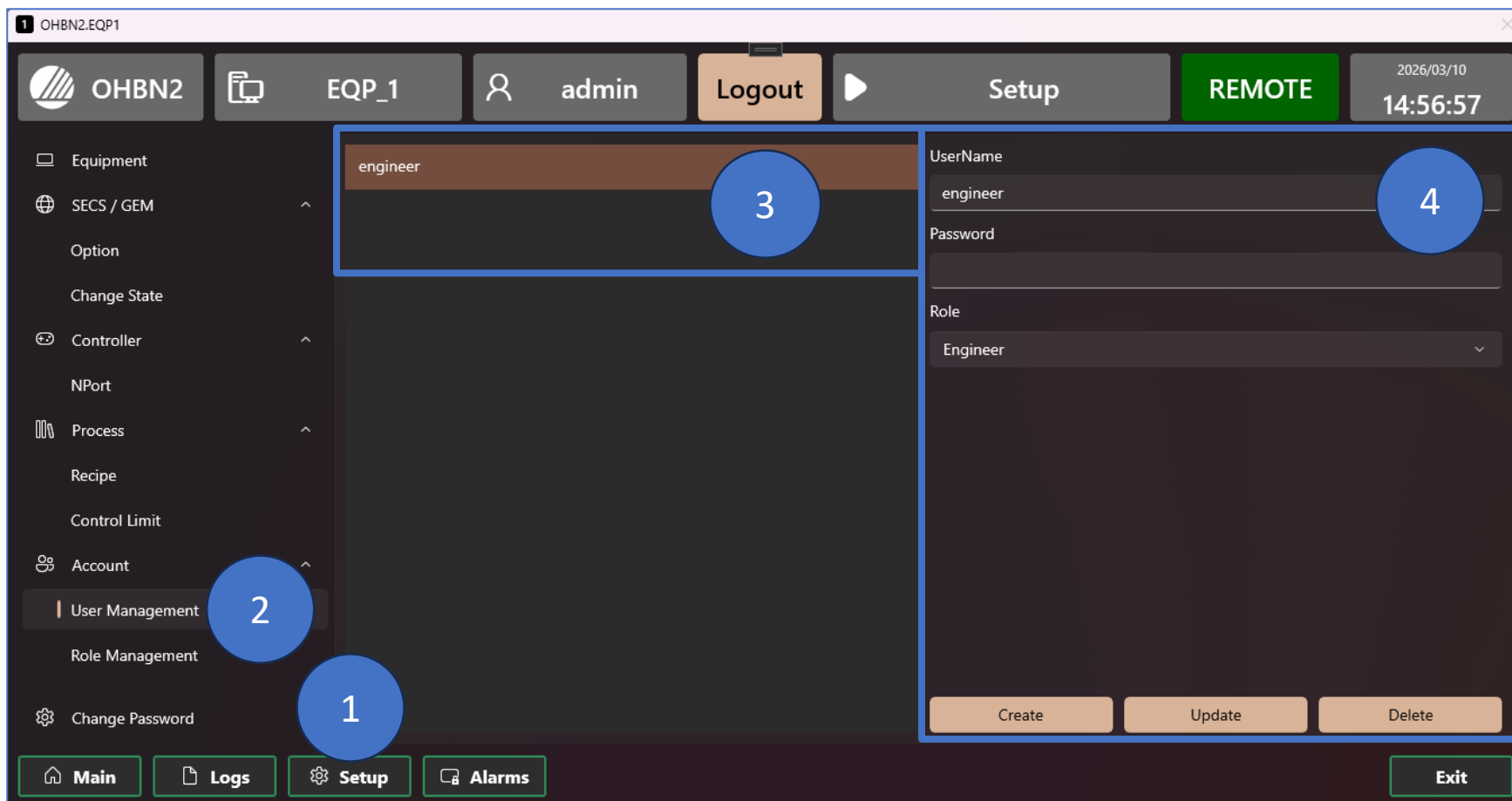
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▼

標題	說明
FOUP On Duration	Placement ON 持續多久視為 FOUP ON
Auto Purge Timeout	FOUP ON 多久後，自動充氣
Low Flow Rate	Recipe 執行完後，切換為小流量，持續充氣。
Economy Flow Rate	Recipe 充氣中，濕度達標後，切換為節能流量。
Humidity Target	濕度目標值
Start Purge Humidity Check Delay	開始充氣後，N 秒後檢查濕度是否達標，不達標發報 Alarm
Rewetting Target	回濕目標值
Rewetting Duration	回濕時間
Rewetting Interval	回濕間隔
Idle Purge Flow Rate	清管流量
Idle Purge Duration	清管時間
Idle Purge Interval	清管間隔
Flow Rate Tolerance Percentage	流量誤差百分比，0 ~ 100 %，超過發報 Alarm。

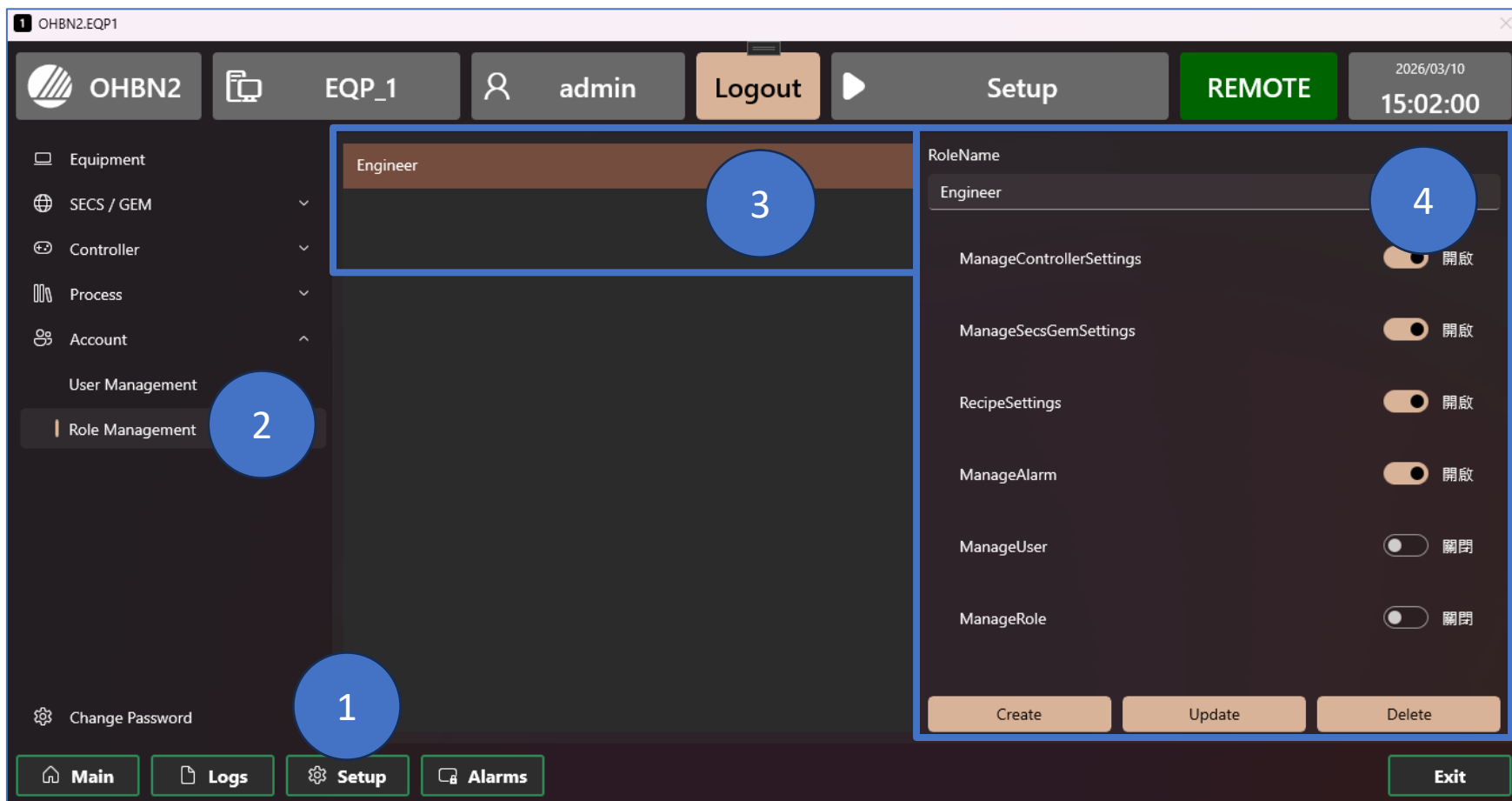
權限設定

使用者管理



標題	說明
1	點選 Setup
2	點選 User Management
3	User 列表
4	User 設定

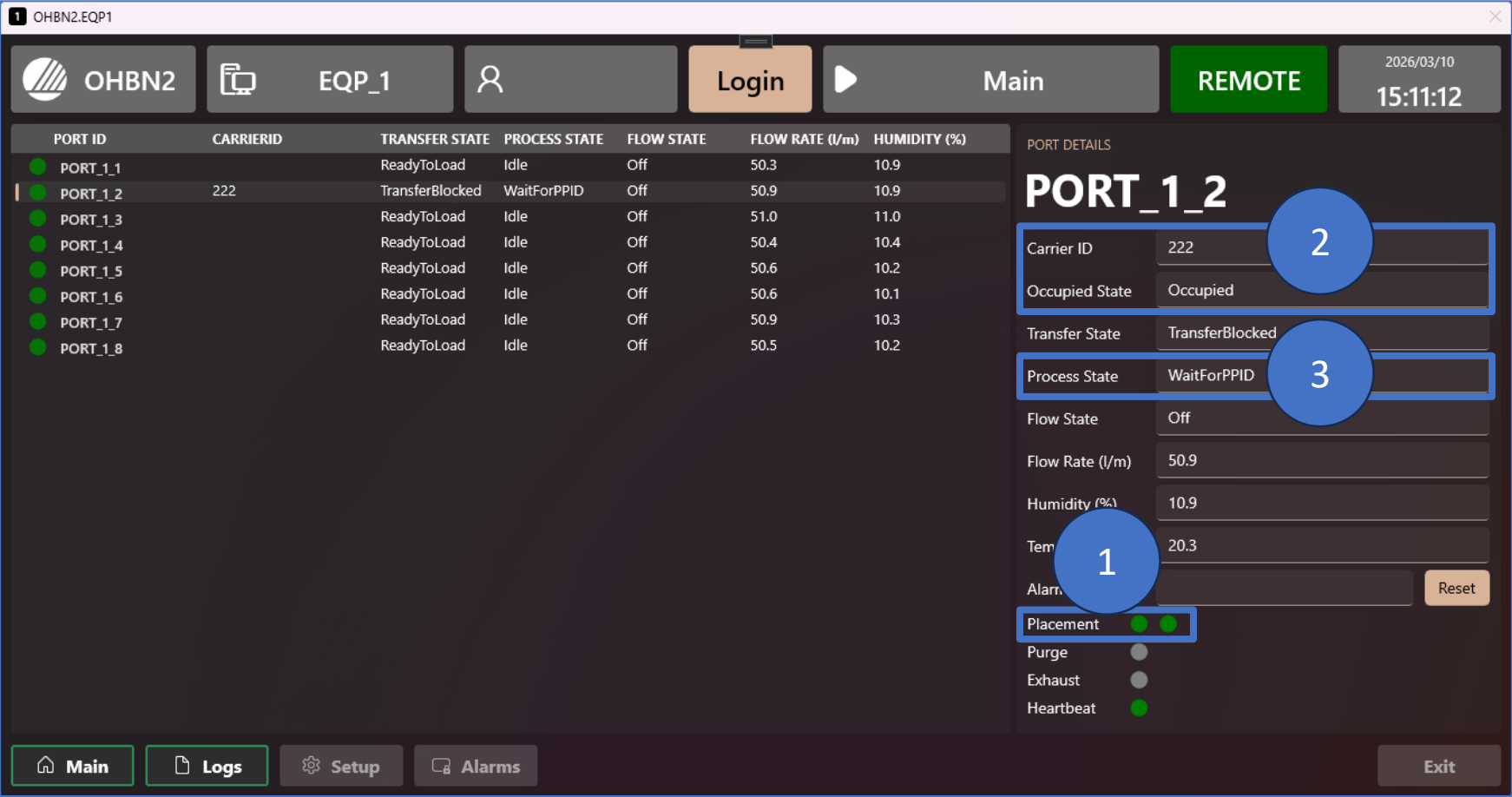
角色管理



標題	說明
1	點選 Setup
2	點選 Role Management
3	Role 列表
4	Role 設定

充氣操作流程

FOUP ON



標題	說明
1	FOUP ON
2	Carrier ID Read
3	等待上位端下 S2F41

自動充氣

OHBN2.EQP1

EQP_1

admin

Logout

Main

REMOTE

2026/03/10 15:22:16

PORT ID	CARRIERID	TRANSFER STATE	PROCESS STATE	FLOW STATE	FLOW RATE (l/m)	HUMIDITY (%)
PORT_1_1		ReadyToUnload	Stopped	Off	50.3	10.7
PORT_1_2	222	ReadyToUnload	Busy	HighFlow	50.3	10.5
PORT_1_3		ReadyToUnload	Stopped	Off	50.1	10.6
PORT_1_4		ReadyToUnload	Stopped	Off	50.1	10.7
PORT_1_5		ReadyToUnload	Stopped	Off	50.5	10.9
PORT_1_6		ReadyToUnload	Stopped	Off	50.1	10.5
PORT_1_7		ReadyToUnload	Stopped	Off	50.1	10.3
PORT_1_8		ReadyToUnload	Stopped	Off	50.8	10.2

PORT DETAILS

PORT_1_2

Carrier ID

222

Occupied State

Occupied

Transfer State

ReadyToUnload

Process State

Busy

Flow State

HighFlow

Flow Rate (l/m)

50.3

Humidity (%)

10.5

Temp

20.8

Ala

Reset

Placem

Purge

Exhaust

Heartbeat

Main

Logs

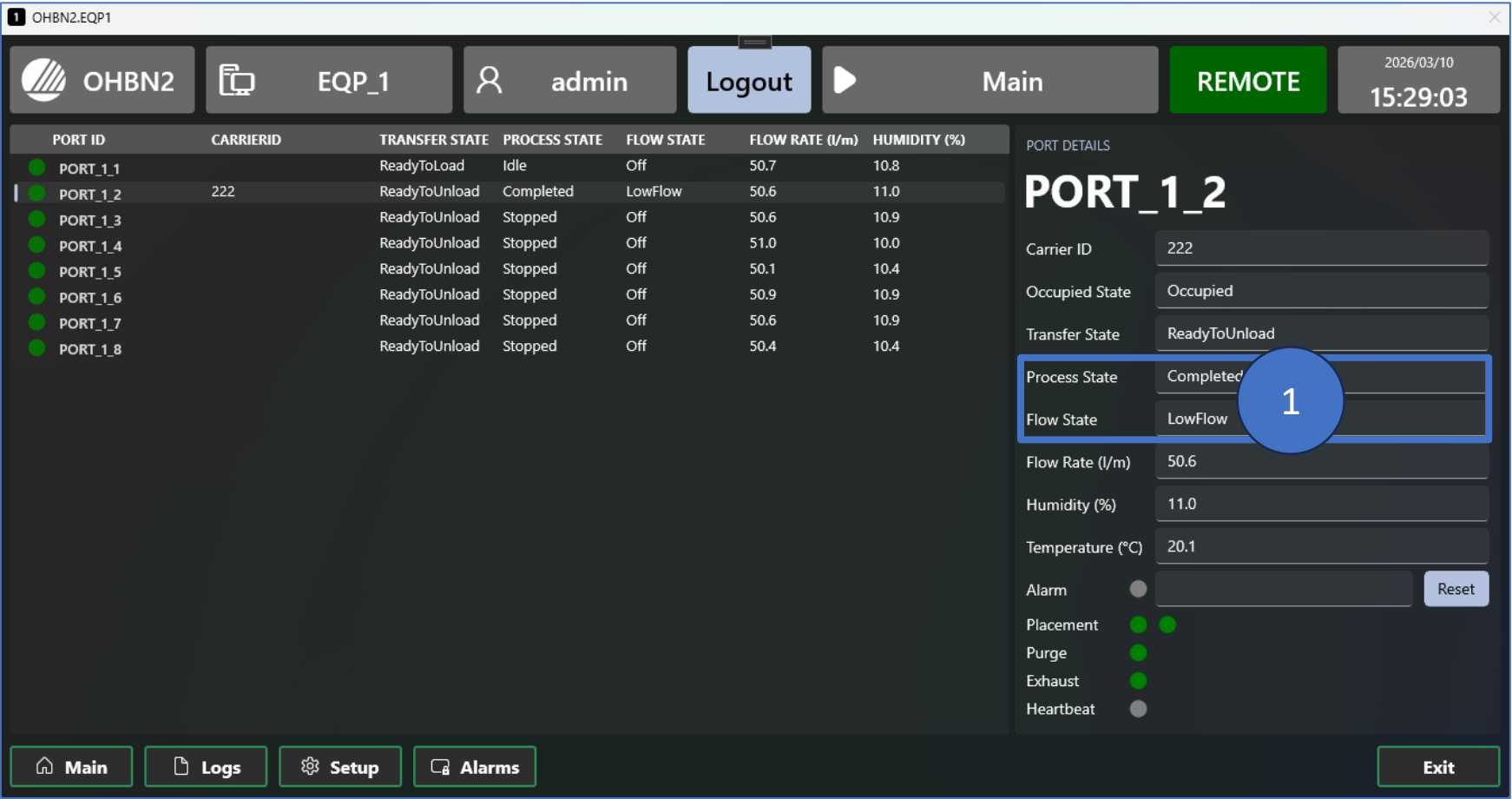
Setup

Alarms

Exit

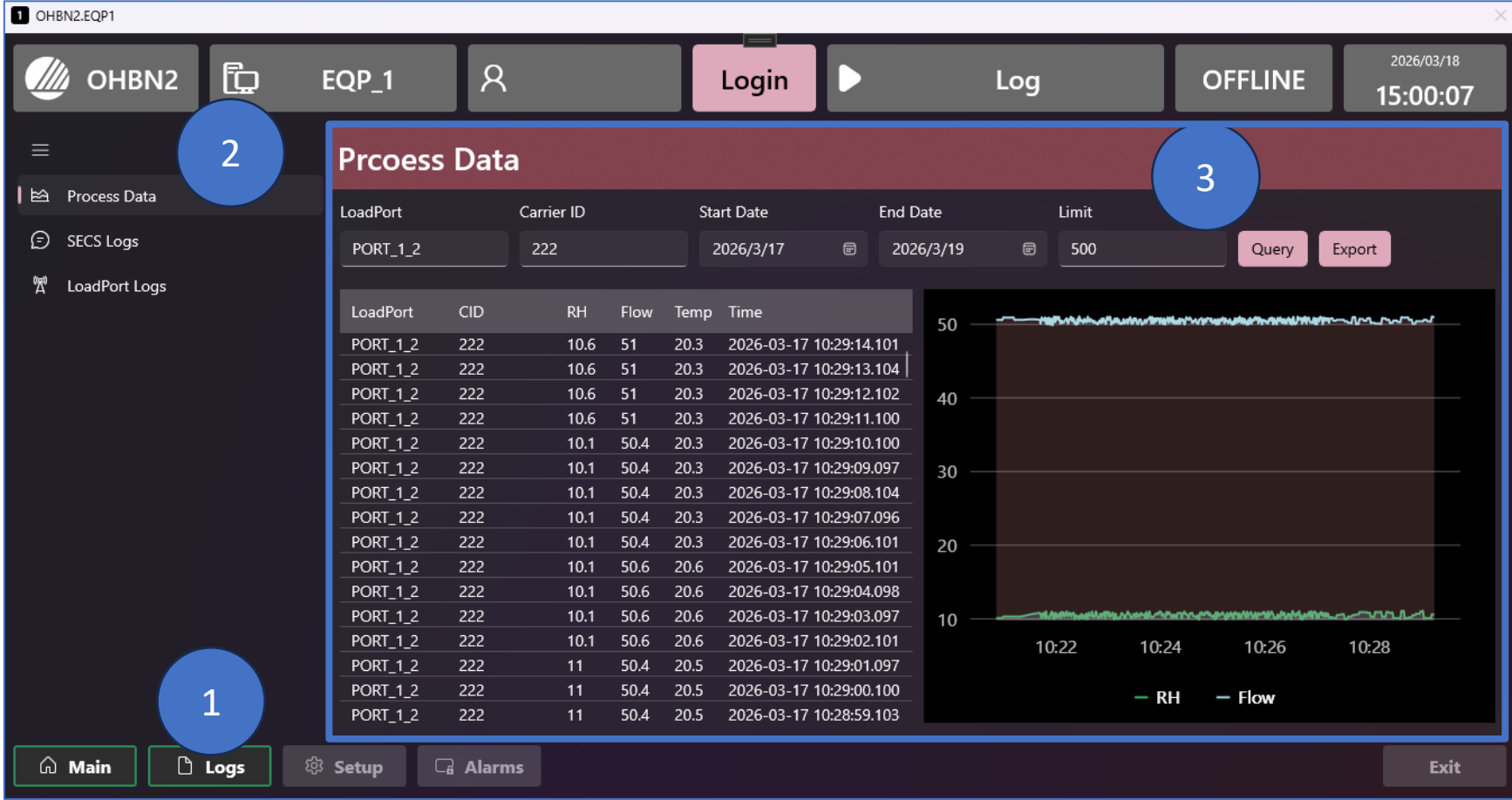
標題	說明
1	FOUP ON 閒置 30 秒後，自動執行 Default Recipe 普通模式：依照 Recipe 充氣完成後，切換為小流量。 節能模式：依照 Recipe 充氣，過程中濕度達到目標值，切換為節能流量。
2	Purge 燈 = 綠燈

充氣完成



標題	說明
1	充氣結束後，切換小流量，直到 FOUP OFF

Process Data



標題	說明
1	點選 Logs
2	點選 Process Data
3	條件篩選完，點選 Query 可查詢充氣資料